MONTREAL.AI RESEARCH REVIEW · v1

AGI ALPHA FLYWHEEL

The Certified Boundary Edition · Executive brief

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A universal answer, available to the next researcher

An executed study has produced an exact mathematical certificate answering Gerry Myerson's documented question about finite congruence partitions. The strongest construction came from the independent conventional challenge. Its proof applies to every interval length and can be checked with ordinary Python, without a model, numerical optimizer or trusted forecast (Myerson 2026).

Partition $1, \ldots, n$ into complete residue classes, each hitting the interval at least twice. Define the **heft** h as the sum of reciprocal moduli. The study proves

$$\boxed{h \geq \frac{17902560}{18258529} = 0.9805039606421744\ldots > 0.98.}$$

This rules out the requested below-0.98 outcome. The proof even allows repeated moduli. It retains complete classes, exact coverage and minimum class size, and makes no claim about singleton classes or two-dimensional covers. The strongest inspected prior general bound is $\sqrt{8} - 2 \approx 0.828427$ (Emanuel 2011). The new constant is not claimed optimal; external expert review and fuller priority assessment remain pending.

What was constructed and checked

A piecewise-linear weight with 21 exact integer heights converts the question into an auditable certificate. Two separately implemented line-arrangement checkers agree on the maximum class weight; an analytic tail covers all larger classes. Assigning each integer its cell integral transfers the inequality to every finite interval length without approximation.

The first successful approach used a degree-20 rational polynomial. It remains independently verified, but its bound is weaker. A fresh workstream, given the public question and ordinary tools without that witness, found the stronger construction. This is an original explicit result relative to inspected sources, obtained with established methods. It does not demonstrate an exclusive algorithm, model superiority or discovery advantage.

The package preserves unsuccessful code searches, an attractive but invalid first polynomial, its exact counterexample, both accepted constructions, frozen configurations, fresh tests and interface corrections. Separate checkers and adversarial tests reduce error risk; they do not constitute external human peer review or proof-assistant formalization.

Useful science; commercial foundation still conditional

Actual customers, revenue and receipts are zero. Complete actual cost and profit are unknown. No profitable enterprise or Operational Alpha is measured.

A complete planning model evaluates reproduction, qualified review and integration of this fixed certificate package, including one year of support. It does not price a promise to solve future open problems. Under explicitly assumed human-resource estimates:

Central conditional arrangement	Amount
Four prospective sponsors at an assumed $6,000 each	$24,000 revenue
Complete project cost, including discovery once	$21,592
Full-project profit before financing and tax	$2,408
Additional unrestricted financing above $5,000 capital	$6,573.20

The central case is profitable but underfunded under its assumptions. Refundable deposits remain restricted. An illustrative 90-day bridge reduces profit to $2,072.63; no lender offer exists. The adverse scenario loses **$19,968.80** and cannot be described as merely underfunded. All fees, demand and labor estimates remain unvalidated.

The supported coordination mechanism is sharing fixed review costs among compatible beneficiaries. An equally capable conventional shared arrangement using the same artifact ties exactly. Specialist extraction remains **HOLD**: its additional modeled $1,042 cost has no demonstrated benefit over giving a competent successor the same ordinary-library asset.

Three horizons and the next decisive step

Near term: a durable answer, exact checkers, clear scope and a preserved correction history. **Expansion:** fresh problems, qualified review, paid acceptance and financing, each tested separately. **Long term:** enabling research instruments and productive infrastructure that leave successors more useful possibilities.

The smallest scientific next step is external mathematical review of scope, proof and priority. Separately, an authorized sponsor interview and actual resource quote can test whether the proposed service is worth buying. No customer contact, minting, trading or public social posting occurred. This is a research-review edition, distinct from a merged canonical release.

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Primary references

Emanuel, Paul Robert. 2011. “Covering Systems.” PhD thesis, Macquarie University. https://doi.org/10.25949/19441907.

Myerson, Gerry. 2026. “Questions on Finite Congruence Covers, in Open Problems Which Might Benefit from Computational Experiments.” https://mathoverflow.net/a/499017.